Anna Bellach, Ph.D.

abellach.biostat@gmail.com | 206 - 771 - 9148 | U.S. permanent resident | German Citizen
linkedin.com/in/anna-bellach-8554a1134 | abellach.github.io

Biostatistician | Clinical Trial Design • Survival Analysis • Statistical Methodology • Consultancy • Regulatory Submissions • Talent Development

Mathematical Statistician / Biostatistician with expertise in survival analysis, recurrent events and competing risks, complex and composite endpoints, clinical trial design, and advanced statistical methodology. Experience collaborating with multidisciplinary teams on multicenter clinical studies, statistical analysis plan development, protocol design, Data and Safety Monitoring Board (DSMB) interactions, and the analysis of complex biomedical data. Track record of methodological research, with publications in leading statistical journals. Experienced in communicating statistical findings to investigators, oversight committees, and scientific stakeholders, and in mentoring postdoctoral fellows and junior statisticians.

Professional Experience

- **Visiting Research Scholar**, University of North Carolina Chapel Hill | 2025 – present
  Conducting methodological research on semiparametric regression models for clustered competing risks data; developing transfer learning approaches for clinical trials with sparse data.
- **Senior Researcher**, German Cancer Research Center, Heidelberg | 2025
  Provided statistical consultancy and methodological leadership for translational oncology research. Contributed to protocol review and SAP development. Mentored PhD students and junior statisticians, providing technical guidance on survival analysis and clinical trial methodology. Conducted methodological research on genomic risk stratification for leukemia trials.
- **Mathematical Statistician / Statistical Methodologist**
  US National Heart, Lung, and Blood Institute (NHLBI / NIH), Bethesda | 2021 – 2024
  Served as Project Statistician for extramural and intramural clinical trials. Provided statistical and methodological expertise for Phase II–III multicenter randomized controlled trials, with emphasis on rigorous study design, endpoint specification, interim monitoring, and benefit–risk evaluation. Contributed to the development of statistical analysis plans (SAPs), protocol review, and regulatory-facing documentation for NHLBI-sponsored multicenter studies. Participated in Data and Safety Monitoring Board (DSMB) activities, supporting evidence-based trial oversight. Mentored postdoctoral fellows and junior statisticians in clinical trial methodology and statistical inference. Completed formal leadership training at NHLBI.
- **Postdoctoral Researcher (Statistical Methodology)**
  Fred Hutch Cancer Center / University of Washington, Yale School of Public Health | 2017 – 2021
  Conducted methodological research on semiparametric regression models for left-truncated and right-censored competing risks data, with application to vaccine efficacy data. Additional research areas: efficient estimation for the case-cohort design, emulated target trials with application to cardiovascular data, personalized biopsy schedules for prostate cancer screening, and metabolic collaborative studies. Coursework in neural networks and deep learning.

Education

PhD **Biostatistics and Bioinformatics**, University of Copenhagen, Denmark
MSc **Mathematics (Diplom)**, University of Freiburg, Germany
BSc **Mathematics (Vordiplom)**, University of Freiburg, Germany
BSc **Economics (Vordiplom)**, University of Freiburg, Germany

Professional Development

- Bridge to Leadership Program, National Institutes of Health (NHLBI)
- Introduction to Bayesian Methods for Clinical Trial Design and Sample Size, ENAR (2022)
- Topics in Causal Inference, Harvard T.H. Chan School of Public Health
- Good Clinical Practice (GCP) Certification, CITI Program

Awards

- **Marie-Curie PhD fellow**, Initial Training Network MEDIASRES (2012 – 2017)
- **SPAC Course Scholarship**, Fred Hutch Cancer Center (2018)
- **Summer Institutes Scholarship**, University of Washington (2018)

Selected Peer-Reviewed Publications and Preprints

A. Methodological Research

1. **Bellach A** & Kosorok MR (2026). Weighted NPMLE for the marginal mean of recurrent events with a competing terminal event. Preprint, arXiv:2605.25934
2. **Bellach A**, Kosorok MR, Gilbert PB & Fine JP (2020). General regression model for the subdistribution of a competing risk under left-truncation and right-censoring. *Biometrika* 107(4): 949–964, DOI: 10.1093/biomet/asaa034
3. **Bellach A**, Kosorok MR, Rüschendorf L & Fine JP (2019). Weighted NPMLE for the subdistribution of a competing risk. *Journal of the American Statistical Association* 114:259–270, DOI: 10.1080/01621459.2017.1401540

B. Collaborative Research

4. Sorror ML, Saber W, Brent RL, Geller N, **Bellach A**, et al. (2024). The Composite Health Risk Assessment Model (CHARM) Predicts Risks of Toxicities, Functional and Cognitive Decline Among Survivors of Allogeneic Hematopoietic Cell Transplantation (allo-HCT): A Prospective BMT-CTN Study 1704. *Blood Advances*, In Press, DOI:10.1182/bloodadvances.2025018340.
5. Sorror ML, Logan B, Saber W, Geller N, **Bellach A**, et al. (2025). Novel Composite Health Assessment Risk Model for Older Allogeneic Transplant Recipients: BMT-CTN 1704. *Blood Advances*, 9 (13): 3268–3280, DOI:10.1182/bloodadvances.2025015793.
6. Cathey B, **Bellach A**, Troendle J, Smith K, Raja N, Osgood S, Kozel B & Levin M (2024). Increased heart rate fragmentation in those with Williams-Beuren syndrome suggests non-autonomic mechanistic contributors to sudden death risk. *AJP-Heart and Circulatory Physiology*, 327(2): H521–H532, DOI:10.1152/ajpheart.00601.2023.

Selected Invited Seminar Talks

1. *Semiparametric regression models for recurrent and terminal events,* St. Jude Children's Research Hospital, Memphis, August 2025.
2. *Semiparametric regression models for recurrent and terminal events,* US National Cancer Institute, Bethesda, November 2023.
3. *Competing risks regression models based on pseudo risk sets*, US National Heart, Lung and Blood Institute, Bethesda (online), July 2021.
4. *Competing risks regression models based on pseudo risk sets*, CBAR at Harvard T.H. Chan School of Public Health, Boston (online), August 2020.

Peer Review Service

Journal of the American Statistical Association, Journal of the Royal Statistical Society: Series B, Biometrics, Annals of Applied Statistics, Lifetime Data Analysis, International Journal of Biostatistics, Statistics in Medicine, Statistical Methods in Medical Research, Statistics and Probability Letters, Statistics in Biosciences, Communications for Statistical Applications and Methods

Programming & Technical Skills

R (advanced): extensive use for simulation studies, methodology development, and clinical data analysis, including publication-grade implementations (JASA, Biometrika). Statistical Simulation & High Performance Computing (HPC): extensive experience designing and executing large-scale simulation studies for methodological validation.

Regulatory & Data Standards: GCP certified; familiar with regulatory principles for clinical trial submissions; actively developing CDISC (SDTM, ADaM) fluency for industry submission workflows.

Languages: English (fluent), German (native speaker)